A Multiscale Spatiotemporal Attention Network for Ground-Based Remote Sensing Cloud Image Sequence Prediction

Feng Zhang, Yingying Cheng, Qiang Hua, Chunru Dong, Yong Zhang, *Senior Member, IEEE*, and Tingdong Wu

***Abstract*— Ground-based cloud image sequence prediction provides valuable insights into cloud motion and meteorological conditions, which are essential for photovoltaic power generation systems. Most existing models are, however, recurrent-based, which is problematic in providing satisfactory forecasting results with rapid speed because these recurrent-based models do not support parallel inference and usually suffer from slow inference speed. A novel recurrent-free deep learning-based framework, called multiscale spatiotemporal attention network (MSTANet) to address the issues is proposed in this study. The MSTANet leverages a multiscale spatiotemporal attention (MSTA) module to extract the multiscale, nonlinear spatiotemporal dependencies from cloud image sequences and uses a multiscale temporal attention (MTA) module to reinforce the temporal dependencies by capturing the high- and low-frequency spatiotemporal fluctuations of clouds. A gated aggregation unit (GAU) to mitigate the ghosting effects that are prevalent in spatiotemporal prediction tasks is introduced to filter the useful context information by integrating the historical information with the updated predictions. Additionally, a multiorder differential divergence regularization term is introduced into the loss function to improve the model's performance by encouraging MSTANet to focus on the evolving trends of the neighborhood of clouds. Experimental results show that the proposed MSTANet outperforms the state-of-the-art (SOTA) prediction methods. It reduces 46% parameters and mean-squared-error (MSE) by 4.31% on the Moving Mnist dataset and reduces 22% parameters with a 1.82% performance improvement on the Folsom dataset compared to the baseline temporal attention unit (TAU). The codes are available at https://github.com/Csorasky/MSTANet.**

***Index Terms*— Deep learning, ground-based cloud images, multiscale attention, recurrent-free, spatiotemporal prediction.**

I. Introduction

PHOTOVOLTAIC power generating systems have received significant attention due to their prominence as a renewable energy source in recent years. Their performance is greatly influenced by the accuracy of solar irradiance prediction [1], which is of great significance to the storage and use of solar power.

Ground-based cloud images provide critical information for predicting solar irradiance, including the sky situations, such as cloud thickness, volume, and cloud distribution; however, these clouds usually variate in appearance, velocity, and size dramatically along time [2]. It is, therefore, crucial to improve the accuracy of forecasting models for ground-based remote sensing cloud image sequences.

Predicting the future ground-based cloud image sequence with high accuracy is quite a difficult task. It has to meet two significant challenges: 1) modeling the nonlinear dynamics of cloud motion trajectories, which are characterized by the splitting and merging of cloud masses, and 2) meeting the necessity of real-time responsiveness for the short-term forecasting since real-time forecasting is crucial for power generation systems [3]. It is, therefore, vital to develop a model to balance the prediction accuracy and inference speed.

Traditional ground-based cloud image sequence prediction methods are mainly linear-based and most of them employed embedded vectors to describe the cloud motion [4], [5], [6], [7], [8]. Typical traditional methods include optical flow [4], similarity maximization [5], [6], and feature tracking [7], [8]. The traditional models have low data requirements and good interpretability; however, they lack the capability of capturing the nonlinear motion of clouds, resulting in imprecise forecasting results.

To tackle this issue, some recent studies framed the ground-based cloud image sequence prediction as a spatiotemporal forecasting problem and built various models by leveraging deep-learning models [9], [10]. The deep learning-based models made prediction by taking historical cloud image sequences as inputs and have achieved great success. Generally, the deep learning-based approaches fall into two categories: recurrent-based [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22] and recurrent-free methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32].

Recurrent-based models employ recurrent neural networks (RNNs) or their variants to capture the temporal dynamics of cloud motion. These methods are more effective in capturing temporal variations and producing improved predictions compared with the traditional approaches. Most of the recurrent-based methods, however, cannot make parallel

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forecasting because of their nature of sequential prediction. In addition, the high computational complexity makes them inappropriate for real-time predictions, especially when the ground-based cloud images are with high resolution.

Alternatively, recurrent-free methods employ convolutional neural networks (CNNs) and attention mechanisms to capture spatiotemporal information from the input sequence. Although the early recurrent-free methods exhibited inferior performance compared to the recurrent-based methods [26], recently proposed recurrent-free approaches demonstrated competitive results by using large kernel attention (LKA) to capture the spatiotemporal features [33]. Most importantly, the recurrent-free methods are capable of balancing the real-time requirements and accuracy of prediction to some extent. The existing recurrent-free methods for ground-based cloud image prediction, however, still face three challenges.

1) The majority of existing recurrent-free models cannot accommodate the variability of multiscaled clouds' motion because they only investigated a single-scale receptive field. To achieve high accuracy, it is recommended to use fine-grained spatiotemporal information for characterizing small-scale cloud formations, while employing coarse-grained information for the clouds with large size.
2) The prevailing recurrent-free prediction methods inadequately capture the intricate spatiotemporal dependencies within image sequence. In practical scenarios, cloud motion exhibits nonstationary movement speeds and frequency fluctuations due to the natural factors such as various wind speed and high-altitude airflow patterns.
3) Existing prediction models with encoders and decoders are prone to producing ghosting effects. In most existing models with encoders and decoders, the downsampling operation is usually employed during the encoding process, which inevitably causes the loss of contextual features. When there is a spatial position offset between the embedding features by the encoder and those features decoded by the decoder, the "ghosting effects" can arise. Although adding residual connections can compensate for some contextual information to some extent, they are still vulnerable to the production of artifacts.

A multiscale spatiotemporal attention network (MSTANet) is proposed to overcome the aforementioned challenges,. MSTANet introduces three main components: 1) a multiscale spatiotemporal attention (MSTA) module, which is designed to capture various cloud motion patterns at different scales, as shown in Fig. 1; 2) a multiscale temporal attention (MTA) module, which is devised to enhance the temporal dependence within a cloud image sequence. It combines both local and global temporal variations, enabling the model to effectively capture both high- and low-frequency fluctuations in a cloud image sequence; and 3) a gated aggregation unit (GAU) that adaptively extracts contextual information and combines it with the subsequent decoder. It helps to mitigate the ghosting effects and improves prediction accuracy by only filtering the useful information during the encoding process. The main contributions of this study are summarized as follows.

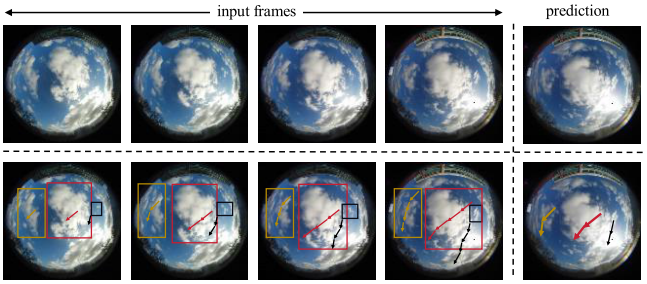


Fig. 1. Illustration of the multiscaled cloud motion with high frequency. The prediction result is obtained using four preceding input cloud frames. The big-sized cloud moves downward to the left (red rectangles) with various velocity (red arrows); the medium-sized cloud moves downward to the left (yellow rectangles) with gradually increasing velocity (yellow arrows); and the small-sized cloud moves significantly fast downward to the bottom (black rectangles, and black arrows with the greatest gradient).

1) A MSTANet that is based on a recurrent-free architecture is introduced for predicting ground-based remote sensing cloud image sequence. It is capable of identifying the clouds with different sizes and capturing their nonlinear motions by employing a multiscale LKA.
2) A MTA module is introduced in MSTANet to enhance a temporal modeling of cloud image sequence. This module improves the model's capability of extracting both high- and low-frequency information by combining local and global temporal variations of cloud motions.
3) A GAU is embedded into the proposed MSTANet. It provides the decoder with context information that is lost during the encoding process, by adaptively retrieving useful historical motion information, which decreases the generation of ghosting effects in predictions.
4) To enable MSTANet to focus on the trend of interframe variation, a multiorder differential divergence is introduced into the loss function. It encourages MSTANet to learn the dynamics of predicted ground-based cloud image sequence.

The rest of the article is organized as follows. Section II reviews the related work of ground-based cloud image sequence prediction models, including traditional methods and deep learning-based methods. Section III provides a detailed implementation of the MSTANet. Extensive experimental results of comparisons on benchmark datasets are provided and discussed in Section IV. Finally, the conclusions and future studies are presented in Section V.

II. Related Studies

The aim of ground-based cloud image sequence prediction is to improve the prediction accuracy of solar irradiance by investigating the motion of clouds in sky, and therefore, enhance the stability of photovoltaic power forecasting systems. The vital challenge is achieving accurate prediction for the future ground-based cloud image sequence with low computational complexity.

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A. Traditional Methods for Ground-Based Cloud Image Sequence Prediction

Most conventional ground-based cloud image sequence prediction approaches are based on linear models. For example, El Jaouhari et al. [51] and Du et al. [34] had proposed a ground-based remote sensing cloud image omnidirectional optical flow tracking method to establish the relationship between the direction and velocity of the cloud motion in a specific area of adjacent frames. To take full advantage of the information on cloud boundary, Chang et al. [35] applied Horn-Schunck optical flow to calculate the motion field of cloud boundary, which improved the forecasting accuracy by embedding more information. Based on the Lukas-Kanade optical flow algorithm and cross correlation, Dissawa et al. [36] proposed a short-term cloud motion tracking method to predict short-term solar irradiance; nevertheless, the approaches brought great computational burden due to dense optical flow calculation for the velocities of every pixel. Later, Wang et al. [37] proposed a model to identify the difference between images by analytically expressions with less computational resources. Jamaly and Kleissl [38] proposed a prediction model by adding quality control to the cross-correlation method and made a cross-spectral analysis to predict cloud motion.

These traditional forecasting methods are simple and displayed good interpretability. The linear-based models, however, cannot handle the complex clouds motion. To capture the nonlinearity of the clouds motion, a variety of deep learning-based techniques have been introduced and achieved remarkable performances [9], [10], [39], [40], [41].

B. Deep Learning-Based Forecasting Methods for Ground-Based Cloud Image Sequence Prediction

The deep learning-based models for ground-based cloud image sequence prediction can be classified into two categories: Recurrent-based and Recurrent-free methods. Recurrent-based methods are the current mainstream, which have undergone extensive investigations, while recurrent-free methods demonstrated competitive performance recent years.

1) Recurrent-Based Methods: Extensive studies have been conducted on recurrent-based prediction models since the groundbreaking work of ConvLSTM, which used 2-D convolution to maintain the spatial motion of sequence [11]. To integrate both temporal and spatial variations, PredRNN [12] was proposed by embedding a zigzag memory flow into ConvLSTM to model short-term spatiotemporal dynamics. In order to mitigate the issue of gradient vanishing, PredRNN++ [13] was proposed by adding a gradient highway unit and a Casual-LSTM module to cascade connect spatial and temporal memories was present. Following the same line, PredRNNv2 [14] introduced a curriculum learning strategy and a memory decoupling loss to improve the performance of PredRNN. Later, Memory In Memory [15] introduced a self-renewed memory module for modeling both nonstationary and stationary sequences. PhyDNet [16] explicitly disentangled PDE dynamics from unknown complementary information via a recurrent physical unit. To model the complex variations of spatiotemporal dynamics, E3D-LSTM [17] built a local motion-aware perception of RNNs and enabled the memory cell to store short-term features by integrating 3-D convolutions into recurrent networks. Recently, a motion-aware unit (MAU) [18] introduced an MAU to capture useful spatiotemporal information. LMC-Memory [19] introduced a long-term motion context memory using memory alignment learning with ConvLSTMs. MotionRNN [20] was proposed to capture the complex spatiotemporal variations of the motion, by introducing a flexible external module for RNN-based predictive models, and significant improvement was achieved. Most recently, a spatiotemporal-aware network (STANet) [10] was proposed to model the cloud's transient variations and cumulative trends by an attention mechanism and achieved state-of-the-art (SOTA) results.

These recurrent-based forecasting methods have achieved great performance at capturing temporal variation; however, due to the sequential nature, they inevitably suffer from the high computational complexity when dealing with high-resolution images, and hinder the models' inference speed.

2) Recurrent-Free Methods: In contrast to the recurrent-based models that employ RNN-CNN to capture the spatiotemporal dependencies and make recurrent forecasting, recurrent-free frameworks leverage CNNs or Transformers to model the spatiotemporal dependencies in cloud image sequences and perform parallelized predictions. The success of the earlier recurrent-free methods heavily relied on the great representation power of CNNs. Typical studies include the deep voxel flow [23] and PredCNN [24]. Deep voxel flow reconstructed a frame by borrowing voxels from near frames and using an autoencoder to learn the voxel flow. PredCNN combined cascade multiplicative units with CNNs to capture the interframe dependencies. For improving prediction accuracy, recent frameworks with knowledge distillation have been proposed by encoding the historical frames and decoding the future semantic segmentation using CNN and distilling [26]. For making predictions at different spatial resolutions, a hierarchical neural model with adversarial and perceptual loss has been proposed as well [30]. To captured multiscale spatiotemporal changes in cloud motion, MSSTNet [32] introduced 3-D convolutions with different kernel sizes. Recent studies have achieved significant improvements in video prediction by UNet [42] and leveraging large kernels [27], [28], [33]. For example, SimVP [27] was proposed for video prediction tasks by simply stacking multiple CNN blocks to learn temporal evolution with a video sequence. Following a similar idea, SimVPv2 [28] introduced a gated spatiotemporal attention module to improve the performance of SimVP, and their ablation study results highlighted the promising potential of the LKA module for extracting spatiotemporal features. Based on this finding, a temporal attention unit (TAU) [29] was proposed by decomposing temporal attention into intra-frame and interframe attention and achieved great performance.

The recurrent-free models take advantage of the parallel prediction and make a balance between inference speed and prediction accuracy. Although the recurrent-free approaches have made significant advancements, unfortunately, they failed

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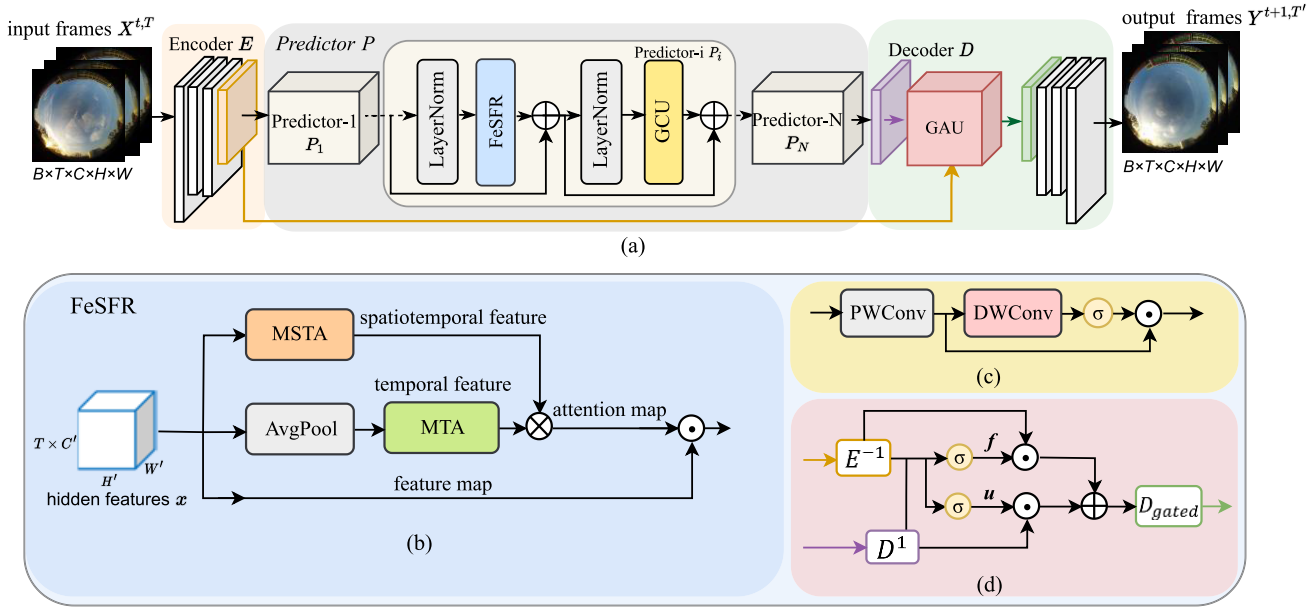


Fig. 2. (a) Overview of the MSTANet. (b) FeSFR that is reinforced by MTA to capture the motion patterns of clouds of various sizes. (c) GCU, which is employed to extract interchannel correlations. (d) GAU that is devised to replenish the context motion information lost during the encoding process. $\oplus, \otimes, \odot$ represents element-wise addition, outer product, and Hadamard product, respectively.

to capture the spatiotemporal variations of the clouds with different sizes. Additionally, most recurrent-free methods have to tackle the prevalent problem of ghosting effects that are caused by the downsampling operations in encoding process; therefore, to break through the dilemmas, an MSTANet is proposed in this study.

III. Method

A. Problem Formulation

Given an input sequence $X^{t,T} = \{x^i\}_{t-T+1}^{t}$ containing T frames up to time t, the goal is to predict the future sequence $Y^{t+1,T'} = \{x^i\}_{t+1}^{T'}$ of T' frames starting from time $t+1$, where $x^i \in \mathbb{R}^{C\times H\times W}$, C is the number of channels, H and W denote the height and width of an image. The input and output is denoted by a tensor, i. e., $\mathcal{X}^{t,T} \in \mathbb{R}^{T\times C\times H\times W}$ and $Y^{t+1,T'} \in \mathbb{R}^{T\times C\times H\times W}$, respectively. The prediction model is a mapping $\mathcal{F}_\Theta: X^{t,T} \rightarrow Y^{t+1,T'}$ with learnable parameters Θ, where $\mathcal{F}_\Theta$ represents a neural network trained to minimize the difference between the predicted cloud images and their ground truths. The parameters Θ^* can be learned by the following equation:

$$\Theta^* = \arg\min_{\Theta} \mathcal{L}\left(\mathcal{F}_\Theta\left(X^{t,T}\right), Y^{t+1,T'}\right). \tag{1}$$

B. Overview of MSTANet

The proposed MSTANet model is explained in detail in this section. The overview of MSTANet is illustrated in Fig. 2(a). It is an encoder-predictor-decoder architecture.

1) Encoder: Encoder E is constructed by alternately stacking downsampling layers and nondownsampling layers. Each downsampling layer is used for feature compression, and the nondownsampling layer is used for feature recombination by a convolution with a stride of one. The kernel size of all layers is set as 3 by default. The encoder can be formulated by the following equation:

$$E_i = \begin{cases} \sigma\left(\text{Conv2d}_{\text{stride}=2}(E_{i-1})\right), & \text{if } 1 \le i \le N_r \text{ and } i \text{ is odd} \\ \sigma\left(\text{Conv2d}_{\text{stride}=1}(E_{i-1})\right), & \text{if } 1 \le i \le N_r \text{ and } i \text{ is even} \end{cases} \tag{2}$$

where the kernel size is set as 3 by default. σ is a sigmoid function, N_r is the number of encoding layers. E_0 represents the input and E_i is the ith layer of encoder. $\text{Conv2d}_{\text{stride}=2}$ represents a convolution with a kernel size of three and a stride of two, and $\text{Conv2d}_{\text{stride}=1}$ represents a convolution with a kernel size of three and a stride of one.

2) Predictor: The predictor stacks six predictor blocks in total. In each block, a frequency-enhanced spatiotemporal feature representation (FeSFR) module, consisting of a MSTA and a MTA, is designed to obtain the motion patterns of multiscaled clouds, as shown in Fig. 2(b); a gated convolutional unit (GCU) is used to extract interchannel correlations of cloud images.

Given an input $x \in \mathbb{R}^{B\times(T\times C')\times H'\times W'}$, the information flow for the ith predictor, P_i, is formulated by the following equations:

$$x' = \text{LN}(x) \tag{3}$$

$$\text{FeSFR}\left(x'\right) = \text{MSTA}\left(x'\right) \otimes \text{MTA}\left(\text{AvgPool}\left(x'\right)\right) \odot x' \tag{4}$$

$$x'' = \left(x + \text{FeSFR}\left(x'\right)\right) \tag{5}$$

$$\text{Predictor}(x) = x'' + \text{GCU}\left(\text{LN}\left(x''\right)\right) \tag{6}$$

where $\text{LN}(\cdot)$ is a layer normalization; $\otimes$ and $\odot$ denotes the outer product and the Hadamard product, respectively.

3) Decoder: The decoder D is constructed by alternately stacking convolutional layers and subpixel convolution [43]. In order to replenish the context information lost during the encoding process, a GAU is introduced to perform a selective aggregation of the embedding features from the encoder and

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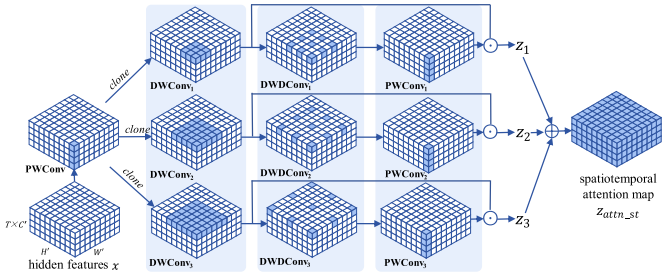


Fig. 3. Details of MSTA. The blue region represents the receptive field of convolutions. A small receptive field tends to capture fine-grained information, and a large receptive field implies long-range dependency. $\odot$ and $\oplus$ represent Hadamard product and element-wise addition.

decoder, where the decoder D_i is formulated by the following equation:

$$D_i = \begin{cases} \mathrm{GAU}\left(\sigma(\mathrm{Conv2d}(D_{i-1})), E^{-1}\right), & \text{if } i = 1 \\ \sigma(\mathrm{Conv2d}(D_{i-1})), & \text{if } 2 \le i \le N_t \text{ and } i \text{ is odd} \\ \sigma(\mathrm{unConv2d}(D_{i-1})), & \text{if } 2 \le i \le N_t \text{ and } i \text{ is even} \end{cases} \tag{7}$$

where σ is an activation function, N_t denotes the number of encoding layers. D_i represent the ith layer of the decoder. Conv2d is a convolution with a kernel size of three and a stride of one; unConv2d represents a subpixel convolution.

C. Frequency-Enhanced Spatiotemporal Feature Representation Module

FeSFR [Fig. 2(b)] is a key component of a predictor. It consists of two parts: 1) an MSTA module that is used to obtain multiscale spatiotemporal features of clouds of various sizes and 2) a MTA, which is introduced to enable the MSTANet to capture both high-frequency and low-frequency temporal variations of clouds motion.

Motivated by the findings that LKA can imitate the attention mechanism with less complexity [33], both MSTA and MTA modules employ the LKA to extract the spatiotemporal features from cloud images. Specifically, LKA divides a $K \times K$ convolution into three parts: 1) a $(2d-1) \times (2d-1)$ depth-wise convolution, i.e., DWConv$(\cdot)$, where d is dilation rate of convolution; 2) a $(K/d) \times (K/d)$ depth-wise dilation convolution with dilation rate d, i.e., DWDConv$(\cdot)$; and 3) a point-wise convolution PWConv$(\cdot)$. Given an embedded feature x, LKA is formulated by the following equation:

$$\mathrm{LKA}(x) = \mathrm{PWConv}(\mathrm{DWDConv}(\mathrm{DWConv}(x))). \tag{8}$$

1) Multiscale Spatiotemporal Attention: The details of MSTA module are shown in Fig. 3. It is devised to capture the multiscale spatiotemporal information within an individual image sequence. MSTA is capable of extracting fine-grained and coarse-grained spatiotemporal information from those small-sized and large-sized clouds by combining multiscale mechanism with LKA.

Specifically, given a hidden feature $x \in \mathbb{R}^{(T\times C')\times H'\times W'}$, it is firstly fed into a point-wise convolution, i.e., PWConv$(\cdot)$, and then cloned parallelly to get three copies, which are fed into three branches to learn their multiscale spatiotemporal features. The features obtained by a depth-wise convolution, i.e., DWConv$(\cdot)$, and the ones obtained by LKA is aggregated to prevent information loss caused by dilation convolution in LKA, which is formulated by (10). Finally, the spatiotemporal attention map $x_{\mathrm{attn_st}}$ is obtained by adding all multiscale spatiotemporal features. The information flow of MSTA is formulated by the following equations:

$$x' = \mathrm{PWConv}(x) \tag{9}$$

$$\mathrm{LKA}_i(x) = \mathrm{PWConv}_i(\mathrm{DWDConv}_i(\mathrm{DWConv}_i(x))) \tag{10}$$

$$z_i = \mathrm{LKA}_i\left(x'\right) \odot \mathrm{DWConv}_i\left(x'\right) \tag{11}$$

$$x_{\mathrm{attn_st}} = \sum_{i=1}^{3} z_i \tag{12}$$

where DWDConv$(\cdot)$ is a depth-wise dilation convolution, $z_i \in \mathbb{R}^{B\times(T\times C')\times H'\times W'}$ is the spatiotemporal attention map of the ith layer, and $x_{\mathrm{attn_st}}$ represents the multiscale spatiotemporal feature map obtained by taking different patterns of cloud motion into account.

The PWConv$(\cdot)$ is used to capture the temporal information by aggregating the temporal features across entire sequence using weighted summation. It tends to capture global temporal features, representing low-frequency temporal variations.

2) MTA: Since the cloud motion is inherently influenced by various speeds and shows different frequencies over time, it is necessary to consider both high- and low-frequency temporal variations to make the prediction more accurate. Because the MSTA mainly focuses on capturing low-frequency temporal features, an additional MTA is introduced to help the MSTA enable MSTANet to capture high-frequency temporal variations and enhance its capability of capturing low-frequency temporal variation without imposing unnecessary computational complexity.

Fig. 4 presents the details of MTA. The local temporal attention is designed to capture high-frequency temporal variations, while the global temporal attention is leveraged to perceive the low-frequency temporal variations.

3) Local Temporal Attention: High-frequency temporal variations typically occur rapidly over short periods. We, therefore, designed local temporal attention to capture the high-frequency temporal variations. Specifically, for a batch of hidden features $x \in \mathbb{R}^{B\times(T\times C')\times H'\times W'}$, average pooling is employed to obtain their embedded feature vectors. After reshaped, the feature vectors are fed into LKA-1-D, which can be regarded as a one-dimension version of LKA, to enhance the MSTANet's capacity to capture high-frequency temporal variations of cloud motion by the following equation:

$$x_{l_\mathrm{Attn}} = \mathrm{LKA\text{-}1D}(\mathrm{AvgPool}(x)) \tag{13}$$

where $x_{l_\mathrm{Attn}} \in \mathbb{R}^{B\times T\times C'}$ denotes the local temporal attention map and AvgPool is average pooling operation.

4) Global Temporal Attention: Low-frequency variations evolve slowly and typically reflect overall cloud motion trends. Hence, a global temporal attention is devised to extract the global temporal features for capturing the low-frequency variations. Specifically, for feature vectors $x_{l_\mathrm{Attn}} \in \mathbb{R}^{B\times T\times C'}$

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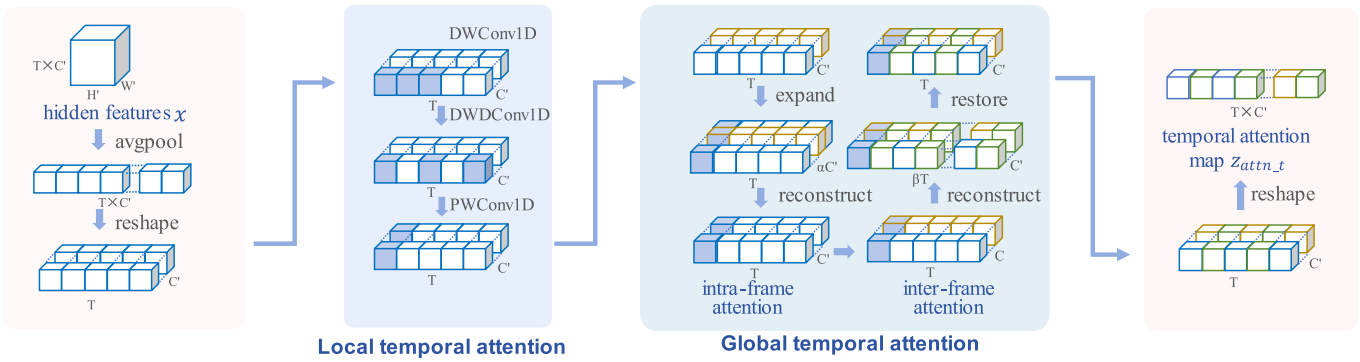


Fig. 4. Details of MTA. The blue region represents the receptive field of convolutions. In order to focus on extracting temporal features, the input data is processed through average pooling and reshaped to obtain tensors that only contain the T and C dimensions. After that, they are fed into an LKA-1-D to get local temporal attention, and then, they are fed into intraframe and interframe attention to get global temporal attention. Finally, the 2-D tensor is transformed into a 1-D tensor in order to merge the temporal features of MTA and the spatiotemporal features of MSTA using the outer product.

obtained from local temporal attention, the global temporal attention consists of intra-frame and interframe attention, which are obtained by expanding and reconstructing along the dimension of channel and time, respectively, where the interframe attention is able to extract the global temporal features for capturing the low-frequency temporal information.

As a result, the time attention matrix $x_{\text{attn_}t}$ that integrates local and global information from the cloud image sequence is obtained. This process is formulated by the following equations:

$$\text{intra-frameAttn} = \text{Conv1d}_{\text{kernel size=1,channels:}(c\to\alpha c\to c)} \tag{14}$$

$$\text{inter-frameAttn} = \text{Conv1d}_{\text{kernel size=1,channels:}(t\to\beta t\to t)} \tag{15}$$

$$x_{g_\text{Attn}} = \text{inter-frameAttn}\left(\text{intra-frameAttn}\left(x_{l_\text{Attn}}\right)\right) \tag{16}$$

$$x_{\text{attn_}t} = \text{reshape}\left(x_{g_\text{Attn}}\right) \tag{17}$$

where $x_{l_\text{Attn}} \in \mathbb{R}^{B\times T\times C'}$ and $x_{g_\text{Attn}} \in \mathbb{R}^{B\times C'\times T}$ denote the local and the global temporal attention map; $x_{\text{attn_}t} \in \mathbb{R}^{B\times T\times C'\times 1\times 1}$ represents the ultimate temporal attention map.

D. Gated Convolutional Unit

GCU is introduced to imitate the function of the multi-layer perceptron (MLP) layer in a vanilla transformer while reducing the complexity of network optimization. It is used to effectively weight the feature maps by using activation functions and depth-wise convolutions instead of matrix multiplication. Empirical results in Section IV show that it is able to reduce approximately 31.53% of parameters while keeping the same accuracy as the MLP layer. The information flow of GCU is formulated by the following equation:

$$x' = \text{PWConv}(x) \odot \sigma(\text{DWConv}(\text{PWConv}(x))) \tag{18}$$

where $x \in \mathbb{R}^{B\times(T\times C')\times H'\times W'}$, σ is an activation function.

E. Gated Aggregation Unit

To mitigate the "ghosting effects," a GAU is devised filter the information provided by the encoder. As shown in Fig. 2(d), the input of the GAU includes the final-layer feature maps from the encoder, i.e., E^{-1}, and the first-layer feature maps from the decoder D^1. The E^{-1} is then fed into a forget gate f, and the D^1 is reweighted by an update gate u. Afterward, they are aggregated to be alternately fed into the stacking convolutional layers and subpixel convolution layers. GAU is formulated by the following equations:

$$\boldsymbol{f} = \sigma\left(W_{ef} * E^{-1} + W_{df} * D^1 + b_f\right) \tag{19}$$

$$\boldsymbol{u} = \sigma\left(W_{eu} * E^{-1} + W_{du} * D^1 + b_u\right) \tag{20}$$

$$D_{\text{gated}} = \left(\boldsymbol{f} \odot E^{-1}\right) \oplus \left(\boldsymbol{u} \odot D^1\right) \tag{21}$$

where $*$ represents a convolution operator; W_{ef}, W_{df}, and b_f are learnable parameters of the forget gate, and W_{eu}, W_{du}, b_u are parameters of the update gate. $\boldsymbol{f}$ represents the forget gate that is used to filter redundant encoded information, while $\boldsymbol{u}$ is the update gate that assigns weights to the decoding information. D_{gated} represents the gate-aggregated information.

F. Multiorder Differential Divergence

To encourage the MSTANet to take the variations of the cloud motion into account, inspired by the TAU [29], a multiorder differential divergence is introduced into the loss function as a regularization term. It is derived by a weighted combination of first-order and second-order differential divergences. The first-order differential divergence is used to reflect the local spatiotemporal trends of the current frame, and the second-order differential divergence is to reflect the smoothness of variations between two adjacent frames.

Given the predicted cloud image $y \in \mathbb{R}^{B\times T\times C\times H\times W}$ and the ground-truth $z \in \mathbb{R}^{B\times T\times C\times H\times W}$, the first-order differential divergence D_1 between the predicted result and the ground-truth are defined by the Kullback–Leibler (KL) divergence based on the following equations:

$$\text{diff}(y_i) = y_{i+1} - y_i \tag{22}$$

$$\text{diff}(z_i) = z_{i+1} - z_i \tag{23}$$

$$\sigma(y)_{t,c,h,w} = \frac{\exp\left(\frac{\text{diff}(y_{t,c,h,w})}{\tau}\right)}{\sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W}\exp\left(\frac{\text{diff}(y_{t,c,h,w})}{\tau}\right)} \tag{24}$$

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$$\sigma(z)_{t,c,h,w} = \frac{\exp\left(\frac{\text{diff}(z_{t,c,h,w})}{\tau}\right)}{\sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W}\exp\left(\frac{\text{diff}(z_{t,c,h,w})}{\tau}\right)} \tag{25}$$

$$D_1(y,z) = D_{\text{KL}}(\sigma(y)|\sigma(z)) = \sum_{i=1}^{T-1}\text{diff}(y_i)\log\frac{\text{diff}(y_i)}{\text{diff}(z_i)} \tag{26}$$

where τ represents the temperature parameter, diff(y) and diff(z) are the first-order differential.

The second-order differential divergence D_2 measures the similarity of the distribution of second-order differentiation, which is defined by the following equation:

$$D_2(y,z) = D_1(\text{diff}(y), \text{diff}(z)). \tag{27}$$

The multiorder divergence is obtained by weighting the combination of first-order and second-order differential divergences, which is defined by the following equation:

$$D_{\text{diff}}(y,z) = \alpha D_1(y,z) + \beta D_2(y,z) \tag{28}$$

where α, β are hyperparameters. Consequently, the overall loss function defined by the following equation is able to focus on the frame-level accuracy and the variation of the cloud motion speed:

$$\mathcal{L} = \sum_{i=1}^{T'}\left\|\hat{\mathcal{Y}} - \mathcal{Y}\right\|^2 + D_{\text{diff}}(y,z). \tag{29}$$

The information flow of the overall MSTANet is shown in the following Algorithm.

Algorithm 1 The Information Flow of the MSTANet

Input: input sequence $X^{t,T} = \{x^i\}_{t-T+1}^{t}$
Output: predictive image sequence $\hat{Y}^{t+1,T'} = \{x^i\}_{t+1}^{T'}$
repeat
  Let layer $i, j, k = 1, loss = 0.0$
  $\mathcal{L} = \sum_{i=1}^{T'}\|\hat{Y} - Y\|^2 + D_{diff}(y,z)$ ($\mathcal{L}$: loss function)
  $E_0 \leftarrow X^{t,T}$ (E: encoder)
  for $i = 1 \rightarrow l$ **do**: (l: the number of encoder)
    if i is odd
    $E_i \leftarrow \sigma(\text{Conv2d}_{stride=1}(E_{i-1}))$
    else
    $E_i \leftarrow \sigma(\text{Conv2d}_{stride=2}(E_{i-1}))$
    $i = i + 1$
  end for
  $P_0 \leftarrow E_l$(P: predictor)
  for $j = 1 \rightarrow m$ **do**: (m: the number of predictor)
    $P_i \leftarrow \text{LayerNorm}(P_{i-1})$
    $P_i \leftarrow \text{MSTA}(P_i) \otimes \text{MTA}(P_i)$
    $P_i \leftarrow \text{LayerNorm}(P_i)$
    $P_i \leftarrow \text{GCU}(P_i)$
    $j = j + 1$
  end for
  $D_0 \leftarrow P_m$(D: decoder)
  for $k = 1 \rightarrow n$ **do**: (n: the number of decoder)
    if i is odd
      $D_i \leftarrow \sigma(\text{Conv2d}_{stride=1}(D_{i-1}))$
    else
      $D_i \leftarrow \sigma(\text{unConv2d}(D_{i-1}))$
    if $k = 1$
      $D_i \leftarrow \text{GAU}(E_l, D_1)$ (GAU: Gated Aggregation Unit)
    $k = k + 1$
  end for
  $loss = \mathcal{L}(\hat{\mathcal{Y}}^{t+1,T'}, \mathcal{Y}^{t+1,T'})$
  $\theta \leftarrow -\nabla_\theta(loss)$
until convergence
return $\hat{Y}^{t+1,T'}$

IV. Experiments

In this section, extensive comparison experiments and ablation studies are conducted on two benchmark datasets, i.e., Moving MNIST [44] and Folsom [45].

A. Evaluation Metrics

The floating-point operations (FLOPs) are used to assess the computational complexity of each model. The inference speed is measured by frames per second (FPS) on a single NVIDIA V100 GPU. The mean-squared-error (MSE) and mean-absolute-error (MAE) are used to evaluate the accuracy of prediction. In addition, the structural similarity index measure (SSIM) and peak signal-to-noise ratio (PSNR) [46] are employed to calculate the similarity between the predicted frames and their ground truth. The value of SSIM ranges from 0 to 1, with higher SSIM and PSNR values indicating greater accuracy in prediction outcomes. All metrics are formally defined by the following equations:

$$\text{MAE} = \frac{\sum_{n=1}^{N}\sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W}\left|x_{n,c,h,w} - y_{n,c,h,w}\right|}{N} \tag{30}$$

$$\text{MSE} = \frac{\sum_{n=1}^{N}\sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(x_{n,c,h,w} - y_{n,c,h,w}\right)^2}{N} \tag{31}$$

$$\text{SSIM}(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{32}$$

$$\text{PSNR} = 20 \times \log_{10}\left(\frac{\text{MAX}_I}{\sqrt{\text{MSE}}}\right) \tag{33}$$

where N denotes the total number of images, $x_{n,c,h,w}$ and $y_{n,c,h,w}$ represents the ground truth and predictions, respectively. μ and σ are the mean and variance, σ_{xy} denotes the covariance of x and y, c_1, and c_2 are two constants. MAX_I is the maximal value of pixels in a cloud image.

B. Experiments Setup

All experiments are conducted using PyTorch and OpenSTL [47] on an NVIDIA A100 GPU. The proposed MSTANet is trained by Adam optimizer [48] with OneCycle learning rate scheduler [49]. Because of the limited computational resources, we train the proposed MSTANet for 200 epochs on Moving MNIST and 100 epochs on the Folsom dataset to evaluate the model's performance. The encoder, predictor, and decoder of MSTANet are stacked by four, eight, and four layers for the Moving MNIST dataset and 4, 12, and 4 layers for the Folsom dataset. Based on our validation results and to make fair comparisons, we follow the settings in references [33], [54] to set the hyperparameters $\{K, d\}$ in LKA as $\{7, 2\}$, $\{14, 3\}$, $\{21, 3\}$, $\{28, 4\}$, and $\{35, 4\}$, where K denotes the kernel size of LKA and d is the dilation rate of LKA. Specifically, three combinations of hyperparameters $\{K, d\}$ for MSTA are tested on the Moving MNIST. The

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TABLE I
PARAMETERS ANALYSIS ON MSTA MODULE WITH DIFFERENT HYPERPARAMETERS $\{K, d\}$

*Set	Params(M)	FLOPs(G)	MSE ↓	MAE ↓	SSIM ↑
set1	24.2	9.4	**45.53**	**119.33**	**0.8927**
set2	23.8	9.2	45.66	120.13	0.8735
set3	24.7	9.5	46.35	123.07	0.8456

*set1: $\{K, d\}$ take the values of {7, 2}, {21, 3} and {35, 4}; set2: $\{K, d\}$ take the values of {7, 2}, {14, 3} and {21, 3}; set3: $\{K, d\}$ take the values of {21, 3}, {28, 4} and {35, 4}.

experimental results are shown in Table I. We observed that when the hyperparameters $\{K, d\}$ take values of set1, the performance of MSTANet is optimal; therefore, in our subsequent experiments, we specified $\{K, d\}$ as {7, 2}, {21, 3}, and {35, 4}.

C. Experimental Results on Moving MNIST

We compared our MSTANet with 11 baselines, including seven recurrent-based methods, i.e., ConvLSTM [11], E3D-LSTM [17], CrevNet [50], PhyDNet [16], MAU [18], PredRNNv2 [14], and DMVFN [53]; and four recurrent-free methods, i.e., SimVP [27], SimVPv2 [28] Earthformer [52], and TAU [29]. Table II shows the comparison results of the 12 methods of Moving MNIST.

Results in Table II demonstrate that the proposed MSTANet achieves best performance in terms of MSE, MAE, SSIM and PSNR. Specifically, compared with DMVFN that achieves the highest FPS, MSTANet reduces MSE and MAE by approximately 80.97% and 60.86%. In addition, compared with DMVFN, MSTANet increases SSIM and PSNR by approximately 16.34% and 43.65%.

The MSE-FLOPs performance of all 12 models is visualized in Fig. 5. It is observed that the proposed MSTANet is positioned closest to the bottom-left corner, indicating that MSTANet is able to balance the prediction performance and computational complexity. Fig. 5 shows that compared with MAU, which is the best-performing recurrent-based method, the FLOPs of MSTANet decreased by approximately 47.19%, and MSE decreased by 12.36%. By comparing with the top-performing recurrent-free baseline method (TAU), the FLOPs are reduced by 41.25%, and MSE is reduced by roughly 4.31%.

These findings imply that MSTANet is competitive with recurrent-based methods in the prediction accuracy by skillfully using the spatiotemporal information extraction capability of recurrent-free approaches.

Fig. 6 illustrates two visualized results for MSTANet on the Moving MNIST dataset. Observations reveal that the target outputs (the second and the fifth rows) manifest scenarios of both digit overlap and separation, despite the absence of such occurrences in the input image sequence. This imposes a heavy burden on models to properly identify the precise moments when digits overlap or disperse by capturing the complex dynamics of movements.

The results in Fig. 6 show that MSTANet is able to accurately distinguish the different movements of the digits and appropriately perceive their movement, even when they will overlap in the future. The visual results provide strong evidence that MSTANet performs well in forecasting the future states of the digits.

D. Experimental Results on Folsom

In this section, we evaluate the proposed model on the Folsom dataset. To verify the performance of the MSTANet, seven SOTA models are taken as baselines, including recurrent-based methods (i.e., ConvLSTM [11], PhyDNet [16], MAU [18], and PredRNNv2 [14]) and recurrent-free methods (SimVP [27], Earthformer [52] and TAU [29]). Table III illustrates the comparison results of eight methods.

Each sequence of the dataset contains 20 images. The training set comprises 4000 sequences, totaling 80 000 images, and the test set consists of 924 sequences, totaling 18 480 images. To avoid training bias caused by imbalanced data partition, fivefold cross validation are used. Each image is resized into 128 × 128 pixels from 1536 × 1536 pixels to reduce the model complexity and demand for GPU resources.

It can be seen from Table III that the MSTANet achieves the best performance compared with the others. Specifically, in terms of MSE, MSTANet performs better than TAU by 1.82%, and it performs significantly better than PhyDNet by 35.49%.

To compare the performance of the eight models in term of both inference speed and prediction accuracy on the Folsom dataset, the MSE-FLOPS performance is visualized in Fig. 7.

The result in Fig. 7 verifies that the MSTANet achieves the best FLOPs-MSE performance (positioned at the bottom-left corner), indicating that MSTANet is capable of balancing both the inference speed and prediction accuracy. As an example, compared with the best-performing recurrent-free baseline (TAU), the MSE of the result provided by MSTANet decreased by 1.82%, while the FLOPs decreased by 25%. Compared with the top-performing recurrent-based baseline (MAU), the FLOPs of MSTANet are reduced by 32.58%, and MSE is reduced by roughly 3.38%.

In order to provide a more comprehensive investigation of model convergence, a closer examination of the training dynamics during the last 30 epochs is visualized in Fig. 8. It can be seen from Fig. 8 that the recurrent-free baselines converge quickly with less fluctuations, compared with the recurrent-based models at the early training stage. Additionally, the training curve of the MSTANet model smoothly converges finally to the lowest value.

Fig. 9 shows the prediction results on the Folsom dataset. The ten frames in the first row are inputs and the second row are ground-truth frames. The results in Fig. 9 demonstrate that the MSTANet's performance outperforms three SOTA prediction approaches. It is observed that the initial frame of prediction results demonstrates a strong similarity to the ground-truth frames, indicating the efficacy of all methods in short-term predictions. As the prediction process progresses, different methods, however, exhibit various degrees of blurriness and inaccuracies, while the MSTANet can generate forecasts that are closest to the ground truth.

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TABLE II
COMPARISONS WITH OTHER METHODS ON THE MOVING MNIST ↓ (OR ↑) INDICATES LOWER (OR HIGHER) IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

		Params(M)	FLOPs(G)	FPS↑	MSE↓	MAE↓	SSIM↑	PSNR↑
Recurrent-based models	ConvLSTM	15.0	56.8	113	29.80	90.64	0.9288	22.10
	E3D-LSTM	51.0	298.9	18	35.97	78.28	0.9320	21.11
	CrevNet	5.0	270.7	10	30.15	86.28	0.9350	22.15
	PhyDNet	**3.1**	15.3	182	28.19	78.64	0.9374	22.62
	MAU	4.5	17.8	201	26.86	78.22	0.9398	22.57
	PredRNNv2	23.9	116.6	52	24.13	73.73	0.9453	**23.21**
	DMVFN	3.5	**0.2**	**1145**	123.67	179.96	0.8140	16.15
Recurrent-free models	SimVP	58.0	19.4	209	32.15	89.05	0.9268	21.84
	SimVPv2	46.8	16.5	282	26.69	77.19	0.9402	22.78
	Earthformer	7.61	34.0	109	46.83	98.45	0.8850	20.36
	TAU	44.7	16.0	283	24.60	71.93	0.9454	23.19
	MSTANet (ours)	24.2	9.4	424	**23.54**	**70.43**	**0.9470**	23.20

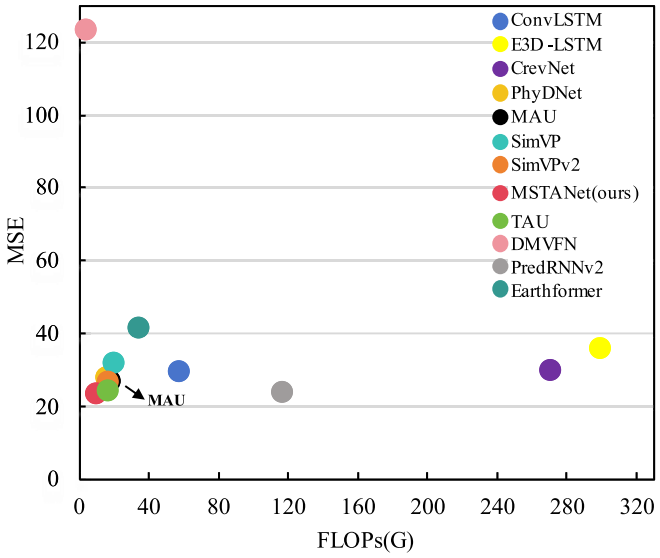


Fig. 5. MSE-FLOPs performance of MSTANet on the Moving MNIST dataset. The better the model, the closer it is to the bottom left corner.

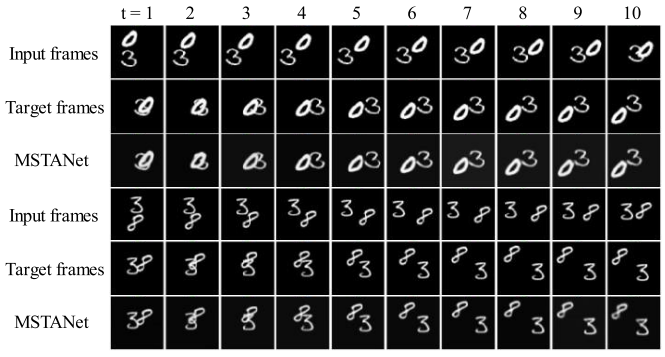


Fig. 6. Examples of predicted results on the Moving MNIST dataset.

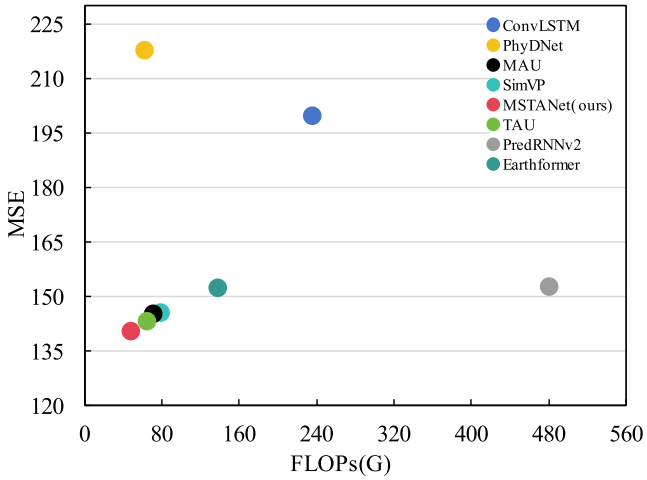


Fig. 7. MSE-FLOPs performance of MSTANet on Folsom dataset. The better the model, the closer it is to the bottom left corner.

TABLE III
COMPARISONS WITH OTHER METHODS ON THE FOLSOM DATASET. ↓ (OR ↑) INDICATES LOWER (OR HIGHER) IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Params(M)	FLOPs(G)	MSE ↓	MAE ↓	SSIM ↑	PSNR ↑
ConvLSTM	15.5	235.0	199.88	1929.33	0.7748	24.96
PredRNNv2	23.9	480.0	152.69	1662.14	0.8102	26.45
PhyDNet	**3.1**	61.4	217.84	2102.95	0.7527	24.17
Earthformer	7.0	138.0	152.49	1610.99	0.8112	26.69
SimVP	58.0	77.9	145.66	1605.26	0.8299	27.08
MAU	7.6	71.2	145.43	1611.52	0.8250	26.78
TAU	44.7	64.0	143.13	1556.80	0.8308	27.26
MSTANet	35.03	**48.0**	**140.52**	**1544.31**	**0.8321**	**27.30**

To clearly demonstrate the effectiveness of the proposed MSTANet, the differences between the ground truth and the predictions-which are defined as the absolute value of the variations between the prediction value and its ground truth in pixels-are calculated and illustrated in Fig. 10. The darker regions indicate smaller differences, indicating that the predic-

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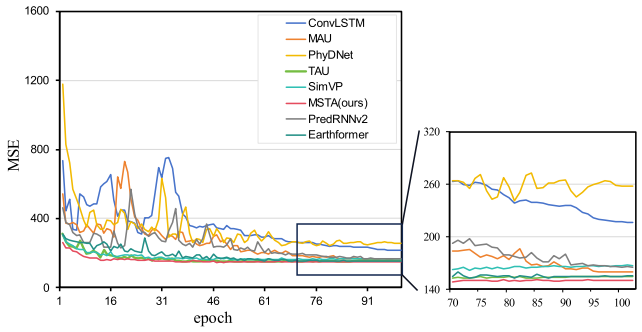


Fig. 8. Training dynamics on the Folsom dataset, x-axis: number of training epochs, y-axis: MSE on the value set.

tions are closer to their ground truth. We compare the error plots of MSTANet, TAU, ConvLSTM, and PhyDNet across ten time steps ($t = 1$ to $t = 10$). Fig. 10 shows that MSTANet makes fewer mistakes compared to other models, highlighting its superior performance on the Folsom dataset.

E. Ablation Studies

In this section, extensive ablation studies have been conducted on Moving MNIST dataset, with the purpose of 1) investigating the importance of each submodule of MSTANet and 2) discovering the influence of the multiorder differential divergence in the loss function on the MSTANet model.

1) Influence of Each Submodule on the Proposed Model: To investigate the effect of each individual module of MSTANet, we gradually remove or substitute each component of MSTANet, including 1) MSTA module, which is used to obtain multiscale spatiotemporal features by employing a multiscale LKA; 2) MTA module, which is introduced to endow the MSTA with a potential for capturing low-frequency temporal variations and enhance the high-frequency temporal variations by a global and a local temporal attention; and 3) GAU module that is employed to capture the correlations of channels in the predictor by effectively adjusting the weight of feature maps via a gating mechanism.

It can be seen from Table IV that the MSTA module has the greatest impact on MSTANet, highlighting the multiscale LKA's capability of capturing complex spatiotemporal information. This result is obtained by replacing the MSTA module with a traditional four-convolution-layers with a kernel size 3, which verifies that the multiscale LKA does cover a larger receptive field than vanilla convolutions, and therefore, achieves superior performance and keeps the inference speed with lower computational complexity.

MTA module brings the performance improvement as well. The reason is that MTA efficiently endows the model with the ability of capturing both low- and high-frequency temporal variations in the sequence. Compared with the individual MSTA module, the MSTA + MTA decreases 5.88% and 2.51% in terms of MSE and MAE, respectively.

In addition, the results in Table IV verified that the GCU module is beneficial to model compression. It decreases approximately 31.53% parameters with a slightly performance improvement. This result is obtained by replacing the MLP layer of vanilla transformer with a Gate Unit.

TABLE IV
ABLATION STUDIES ON MODULES OF MSTANET

MSTA	MTA	GAU	GCU	Params(M)	FLOPs(G)	MSE↓	MAE↓	SSIM↑
√				34.0	13.3	53.24	133.84	0.8617
	√			29.6	11.8	56.30	137.47	0.8570
√	√			35.2	13.3	50.11	130.48	0.8657
√	√	√		35.3	12.3	49.30	130.27	0.8825
√	√		√	**24.1**	**10.4**	46.49	119.78	0.8789
√	√	√	√	24.2	9.4	**45.53**	**119.33**	**0.8927**

TABLE V
PARAMETERS ANALYSIS RESULTS ON HYPERPARAMETERS $\{K, d\}$

$\{7, 2\}$	$\{21, 3\}$	$\{35, 4\}$	Params(M)	FLOPs(G)	MSE ↓	MAE ↓	SSIM ↑
√			18.6	7.9	47.26	124.74	0.8299
	√		19.2	8.0	46.08	123.14	0.8566
		√	21.6	8.7	45.96	122.64	0.8648
√	√	√	24.2	9.4	45.53	119.33	0.8927

TABLE VI
ABLATION STUDIES ON MULTIORDER DIFFERENTIAL DIVERGENCE

	MSE↓	MAE↓	SSIM↑
MSE	45.59	119.52	0.8855
MSE + γD_1	45.31	115.10	0.8872
MSE + D_{diff}	**44.81**	**113.98**	**0.8907**

It is observed from Table IV that the SSIM has been increased by 1.57%, compared with the MSTANet w/o GAU module, which indicates that the generation ratio of ghosting effects has been decreased by GAU. A qualitative comparison experiment is conducted to intuitively verify the effectiveness of GAU in reducing the ghosting effects of the predictions. Because of the fact that removing MTA and MSTA would result in a lack of functional prediction modules in MSTANet, the MTA and MSTA modules are kept. As shown in Fig. 11, MSTANet with GAU makes better predictions compared with its counterpart, indicating that the GAU module can mitigate ghosting effects that commonly occur in ground-based remote sensing cloud image sequence prediction tasks.

Additionally, we perform an ablation study to evaluate the impact of multiscale on the MSTANet model. The results in Table V demonstrate that applying multiscale LKA can significantly increases the prediction accuracy, which implies that the LKA with various receptive fields can capture the motions of clouds of varying sizes, allowing the model to incorporate both local and global spatiotemporal data.

2) Impact of the Regularization Term in Loss Function on the Proposed Model: An extra ablation study is conducted to investigate the impact of the newly defined loss function on the proposed MSTANet. The results are presented in Table VI, where the D_{diff} denotes the multiorder differential divergences, which is a linear combination of first-order and second-order

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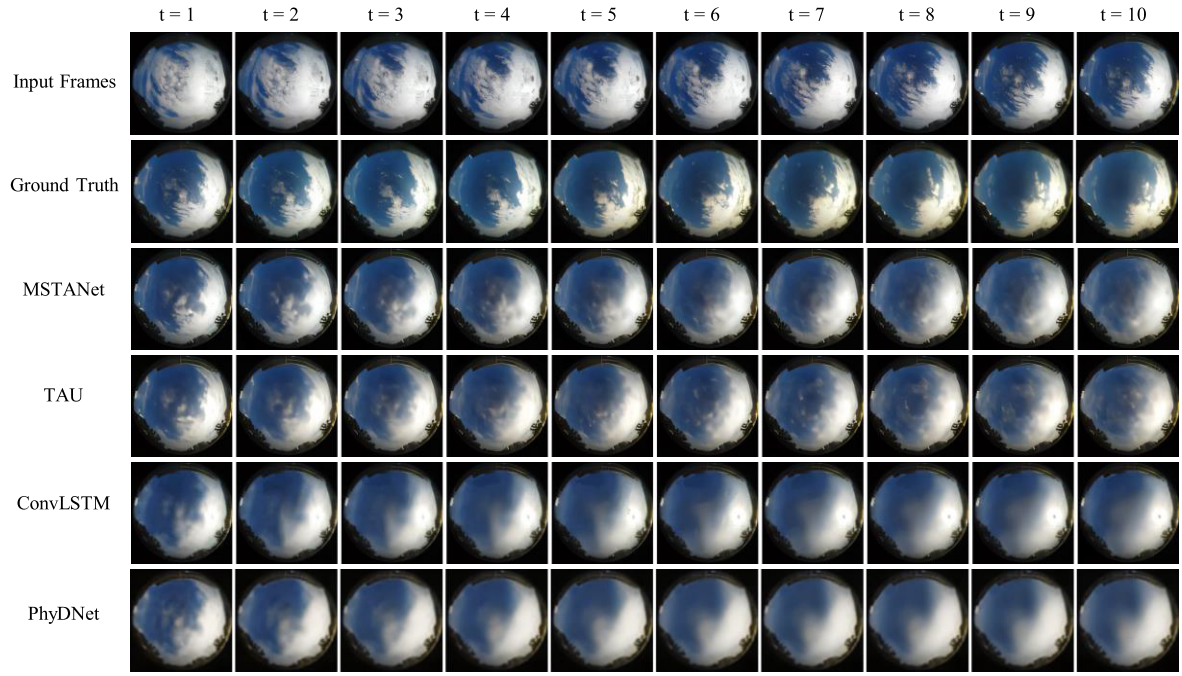


Fig. 9. Examples of prediction results on the Folsom dataset.

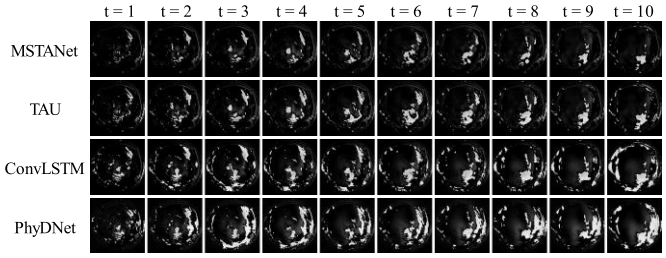


Fig. 10. Set of visualizations depicting the difference between actual values and predictions. The darker the regions in the image, the closer the predicted result is to the ground truth.



Fig. 11. Demonstration sample to highlight the influence of GAU. (a) Ground-truth. (b) Predicted cloud images by MSTANet without GAU. (c) Predicted cloud images by MSTANet with GAU. Red rectangles denote regions by MSTANet without GAU, while black rectangles indicate areas where predictions are notably improved with the inclusion of GAU.

differential divergences. $D_{\text{diff}} = \alpha D_1(y, z) + \beta D_2(y, z)$, here we take $\alpha = 0.1$, $\beta = 0.1$ and $\gamma = 0.2$.

Table VI illustrates that optimizing both the multiorder differential divergence and MSE can produce best performance. The reason is straightforward. Combining the first-order differential divergence term D_1 with vanilla MSE can encourage MSTANet to focus on the trends of the nearest neighbors of the current frame. In addition, Table VI shows that MSTANet keeps performance improvement when the second-order differential divergence D_2 is added into the loss function. This finding implies that D_2 is also beneficial, as it enhances the spatiotemporal scope within which the model is capable of discriminating the patterns of cloud motion.

V. Conclusion

In this study, an MSTANet is proposed for ground-based cloud image sequence prediction. It integrates an MSTA that captures multiscale spatiotemporal dependencies within cloud motion, an MTA that models both high-frequency and low-frequency spatiotemporal variations of cloud motion, and a GAU that compensates for the information that is lost during the encoding process. Additionally, a multiorder differential divergence loss function is introduced into the traditional MSE to improve the model performance. Extensive experimental results on benchmark datasets have shown the MSTANet outperforms the SOTA baselines with significantly decreased computational complexity.

Although the MSTANet reduced computational complexity, it inevitably comes with a cost of compromising essential information that could potentially enhance prediction accuracy. For addressing this issue, the recently proposed dynamic convolutional attention mechanisms may offer promising solutions. Future studies will try to integrate dynamic convolutional attention mechanisms into our prediction model.

Acknowledgment

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